

# Lectures 5b & 6: Superconducting Qubits

## Cooper Pair Box, Transmon, Flux Qubits, Circuit QED, and Decoherence

Comprehensive Study Guide with Full Derivations

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# 1. The Cooper Pair Box (CPB) Qubit

## 1.1 Physical Setup

The Cooper Pair Box is the simplest superconducting qubit. It consists of a small superconducting **island** connected to a large grounded **reservoir** (“bulk”) through a Josephson junction. A gate capacitor  $C_g$  and gate voltage  $V_g$  allow tuning an offset charge on the island.

The key degree of freedom is  $n$ : the number of *excess* Cooper pairs on the island relative to some reference. Cooper pairs tunnel on and off the island through the Josephson junction.

## 1.2 The CPB Hamiltonian — Full Derivation

### Deriving the CPB Hamiltonian with gate charge

The island has total capacitance  $C_\Sigma = C_J + C_g$  (junction capacitance plus gate capacitance). The total charge on the island is  $Q = 2e\hat{n}$ , but the gate voltage induces an **offset charge**  $Q_g = C_g V_g$ . In units of Cooper pairs:  $n_g = C_g V_g / (2e)$ .

The electrostatic energy of the island with  $n$  excess Cooper pairs, offset by the gate-induced polarization charge  $n_g$ :

$$H_{\text{el}} = \frac{(2e)^2}{2C_\Sigma} (\hat{n} - n_g)^2 = 4E_C (\hat{n} - n_g)^2 \quad (1)$$

where  $E_C = e^2/(2C_\Sigma)$  is the single-electron charging energy. The factor of 4 arises because Cooper pairs carry charge  $2e$ :  $(2e)^2/(2C_\Sigma) = 4 \cdot e^2/(2C_\Sigma) = 4E_C$ .

The Josephson tunneling energy, as derived in Lecture 5:

$$H_J = -E_J \cos \hat{\delta} \quad (2)$$

The total CPB Hamiltonian is:

$$\hat{H}_{\text{CPB}} = 4E_C (\hat{n} - n_g)^2 - E_J \cos \hat{\delta} \quad (3)$$

## 1.3 The CPB Hamiltonian in the Charge Basis

In the charge (number) basis  $\{|n\rangle\}$ ,  $\hat{n}$  is diagonal:  $\hat{n}|n\rangle = n|n\rangle$ . Using  $\cos \hat{\delta} = \frac{1}{2} \sum_n (|n\rangle\langle n+1| + |n+1\rangle\langle n|)$  (from the Fourier relation between phase and number states):

$$\hat{H}_{\text{CPB}} = \sum_n \left[ 4E_C (n - n_g)^2 |n\rangle\langle n| - \frac{E_J}{2} (|n\rangle\langle n+1| + |n+1\rangle\langle n|) \right] \quad (4)$$

This is a **tridiagonal matrix** in the charge basis — the diagonal elements  $4E_C (n - n_g)^2$  are the charging energies of each charge state, and the off-diagonal elements  $-E_J/2$  are the tunneling couplings between adjacent charge states:

$$\mathbf{H} = \begin{pmatrix} \ddots & & & & \\ & 4E_C(-1 - n_g)^2 & -E_J/2 & & \\ & -E_J/2 & 4E_C(0 - n_g)^2 & -E_J/2 & \\ & & -E_J/2 & 4E_C(1 - n_g)^2 & \\ & & & & \ddots \end{pmatrix} \quad (5)$$

Diagonalizing this matrix numerically gives the energy levels as a function of  $n_g$ .

## 1.4 The CPB Hamiltonian in the Phase Basis

### Phase-basis representation and Mathieu equation

The (normalized) phase states are  $|\delta\rangle = \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} e^{in\delta} |n\rangle$ , with  $\delta \in [0, 2\pi)$ . Completeness and orthogonality:

$$\langle n|n'\rangle = \delta_{nn'}, \quad \sum_n |n\rangle\langle n| = \mathbf{1}, \quad \langle \delta|\delta'\rangle = \delta(\delta - \delta'), \quad \int_0^{2\pi} d\delta |\delta\rangle\langle \delta| = \mathbf{1} \quad (6)$$

In the phase basis, the number operator becomes a derivative:  $\hat{n} \rightarrow \frac{id}{d\delta}$  (acting on the wave function  $\Psi(\delta) = \langle \delta|\Psi\rangle$ ). This follows from  $\hat{n}|\delta\rangle = \frac{1}{\sqrt{2\pi}} \sum_n n e^{in\delta} |n\rangle = -i \frac{d}{d\delta} |\delta\rangle$ , so when sandwiched:  $\langle \delta|\hat{n}|\Psi\rangle = -i \frac{d}{d\delta} \langle \delta|\Psi\rangle$ .

Wait — the sign:  $\hat{n}|\delta\rangle = \frac{1}{\sqrt{2\pi}} \sum_n n e^{in\delta} |n\rangle$ . And  $\frac{d}{d\delta} |\delta\rangle = \frac{1}{\sqrt{2\pi}} \sum_n (in) e^{in\delta} |n\rangle = i \hat{n} |\delta\rangle$ . So  $\hat{n}|\delta\rangle = \frac{1}{i} \frac{d}{d\delta} |\delta\rangle = -i \frac{d}{d\delta} |\delta\rangle$ , giving  $\hat{n} \rightarrow -i \frac{d}{d\delta}$  when acting to the right. But in the lecture, it's written as  $\hat{n} \rightarrow \frac{id}{d\delta}$  acting on  $\langle \delta|\Psi\rangle$ . Let's be careful:

$\langle \delta|\hat{n}|\Psi\rangle = ?$  Insert completeness  $\sum_n |n\rangle\langle n|$ :  $\langle \delta|\hat{n}|\Psi\rangle = \sum_n n \langle \delta|n\rangle \langle n|\Psi\rangle = \sum_n n \frac{e^{-in\delta}}{\sqrt{2\pi}} \langle n|\Psi\rangle$ . Meanwhile  $-i \frac{d}{d\delta} \langle \delta|\Psi\rangle = -i \frac{d}{d\delta} \sum_n \frac{e^{-in\delta}}{\sqrt{2\pi}} \langle n|\Psi\rangle = -i \sum_n (-in) \frac{e^{-in\delta}}{\sqrt{2\pi}} \langle n|\Psi\rangle = \sum_n n \frac{e^{-in\delta}}{\sqrt{2\pi}} \langle n|\Psi\rangle$ . ✓

So indeed  $\hat{n} \rightarrow -i \frac{d}{d\delta}$  in the  $\langle \delta|$  representation, which can be written  $\hat{n} = \frac{id}{d\delta}$  with the convention in the slides.

The Schrödinger equation  $\hat{H}|\Psi\rangle = E|\Psi\rangle$  in the phase basis becomes:

$$\left[ 4E_C \left( \frac{id}{d\delta} - n_g \right)^2 - E_J \cos \delta \right] \Psi(\delta) = E \Psi(\delta) \quad (7)$$

With the substitution  $\delta = 2u$  and defining  $a = -E/(2E_C)$ ,  $q = -E_J/(2E_C)$ , this maps to the **Mathieu equation**:

$$\frac{d^2 \Psi}{du^2} + (a - 2q \cos 2u) \Psi = 0 \quad (8)$$

The exact solutions are **Mathieu functions**  $me_{-2m} \left( -\frac{E_J}{2E_C}, \frac{\delta}{2} \right)$ , with eigenvalues given by the Mathieu characteristic values.

### Physical Insight

The **charge basis is convenient** when  $E_J \ll E_C$  (only a few charge states matter, so truncation to a small matrix works well). **The phase basis is convenient** when  $E_J \gg E_C$  (the particle is localized in the cosine well, and the wave function is smooth in  $\delta$ ). In the intermediate regime  $E_J \sim E_C$ , numerical diagonalization in either basis works.

## 2. Energy Spectrum and the Two-State Approximation

### 2.1 Energy Levels vs. Gate Charge

The energy levels  $E_m(n_g)$  are periodic in  $n_g$  with period 1 (adding one Cooper pair to the island is physically equivalent to shifting  $n_g \rightarrow n_g + 1$ ). The key features:

At  $n_g = 0$ : the ground state is  $|n = 0\rangle$  (zero excess Cooper pairs). The charging energy parabolas  $4E_C(n - n_g)^2$  have their minimum at  $n = 0$ , and the first excited state is  $|n = \pm 1\rangle$  at energy  $4E_C$ .

At  $n_g = 0.5$ : the states  $|n = 0\rangle$  and  $|n = 1\rangle$  are **degenerate** — both cost charging energy  $4E_C(0.5)^2 = E_C$ . The Josephson coupling  $-E_J/2$  lifts this degeneracy, creating an **avoided crossing** with splitting  $E_J$ .

### 2.2 Two-State Approximation at the Charge Degeneracy Point

#### Reducing to a qubit Hamiltonian

At  $n_g \approx 0.5$ , only states  $|n = 0\rangle$  and  $|n = 1\rangle$  are relevant (all others are far away in energy). In the subspace  $\{|0\rangle, |1\rangle\}$ :

$$\hat{n} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1 - \hat{\sigma}_z}{2}, \quad \cos \hat{\delta} \rightarrow \frac{\hat{\sigma}_x}{2} \quad (9)$$

(The second relation comes from  $\cos \hat{\delta} = \frac{1}{2}(|0\rangle\langle 1| + |1\rangle\langle 0|) = \frac{1}{2}\sigma_x$  in this 2D subspace.) Substituting into the CPB Hamiltonian:

$$H_2 = 4E_C \left( \frac{1 - \sigma_z}{2} - n_g \right)^2 - \frac{E_J}{2} \sigma_x \quad (10)$$

$$= 4E_C \left( \frac{1}{2} - n_g - \frac{\sigma_z}{2} \right)^2 - \frac{E_J}{2} \sigma_x \quad (11)$$

Expanding and dropping the constant (identity) terms:

$$\boxed{H_2 = -E_C(1 - 2n_g)\sigma_z - \frac{E_J}{2}\sigma_x} \quad (12)$$

At the **sweet spot**  $n_g = 1/2$ : the  $\sigma_z$  term vanishes, and  $H_2 = -\frac{E_J}{2}\sigma_x$ . Performing a basis rotation  $\sigma_x \rightarrow \sigma_z$  (i.e., diagonalizing):

$$\boxed{\tilde{H}_2 = -\frac{E_J}{2}\sigma_z} \quad (13)$$

This is a pure spin-1/2 system with level splitting  $E_J$ . At  $n_g = 1/2$ , the qubit frequency is  $\omega_q = E_J/\hbar$ .

#### Critical Point

The  $n_g = 1/2$  sweet spot is where the qubit is first-order insensitive to charge noise:  $\partial\omega_q/\partial n_g = 0$ . But *second-order* sensitivity remains. Since charge noise  $\delta n_g$  is ubiquitous (due to charge traps in the substrate and oxide), the CPB has  $T_2 \sim 100$  ns — far too short for useful quantum computing. This is the fundamental problem that the transmon solves.

### 3. Wave Functions in the CPB and Transmon Regimes

Understanding the wave functions gives deep physical insight into what changes as  $E_J/E_C$  increases.

#### 3.1 CPB Regime: $E_J \ll E_C$

Charging energy dominates  $\Rightarrow$  the system “wants” to have a well-defined number of Cooper pairs.

At  $n_g = 0$ : The ground state is essentially  $|\psi_0\rangle \approx |n = 0\rangle$  — a single charge state. In the phase representation,  $\Psi_0(\delta) = \langle \delta | 0 \rangle = 1/\sqrt{2\pi}$ , a constant (flat) function — the phase is *completely uncertain*. The excited state is  $|\psi_1\rangle \approx |n = 1\rangle$ , with  $\Psi_1(\delta) = e^{i\delta}/\sqrt{2\pi}$ .

At  $n_g = 0.5$ : States  $|0\rangle$  and  $|1\rangle$  are degenerate; the Josephson coupling hybridizes them into:

$$|\psi_0\rangle \approx \frac{|n=0\rangle + |n=1\rangle}{\sqrt{2}} \Rightarrow \Psi_0(\delta) = \frac{1 + e^{i\delta}}{\sqrt{4\pi}} \propto \cos(\delta/2) \quad (14)$$

$$|\psi_1\rangle \approx \frac{|n=0\rangle - |n=1\rangle}{\sqrt{2}} \Rightarrow \Psi_1(\delta) = \frac{1 - e^{i\delta}}{\sqrt{4\pi}} \propto \sin(\delta/2) \quad (15)$$

In the phase representation,  $|\Psi_0(\delta)|^2$  peaks at  $\delta = 0$  (in phase) while  $|\Psi_1(\delta)|^2$  peaks at  $\delta = \pm\pi$  (out of phase). But both distributions are broad — the phase is still poorly defined.

#### 3.2 Transmon Regime: $E_J \gg E_C$

Josephson energy dominates  $\Rightarrow$  the system “wants” well-defined phase, localized near  $\delta = 0$  (the bottom of the cosine well).

The ground state in the *number basis* spreads over *many* charge states:  $\psi_0(n)$  is a broad, approximately *Gaussian* envelope centered at  $n \approx 0$ . In the *phase basis*,  $\Psi_0(\delta)$  is a narrow *Gaussian* peaked at  $\delta = 0$  — looking exactly like a harmonic oscillator ground state in the cosine well.

The first excited state  $\Psi_1(\delta)$  has one node at  $\delta = 0$ , the second  $\Psi_2(\delta)$  has two nodes — just like the wave functions of a particle in a potential well.

#### Key Result

The progression from CPB to transmon is a smooth crossover:

|                          | CPB ( $E_J/E_C \ll 1$ )     | Transmon ( $E_J/E_C \gg 1$ )     |
|--------------------------|-----------------------------|----------------------------------|
| Good quantum number      | Charge $n$                  | Phase $\delta$                   |
| $\psi_0$ in charge basis | Sharp peak at one $n$       | Broad Gaussian over many $n$     |
| $\psi_0$ in phase basis  | Flat (delocalized)          | Narrow Gaussian at $\delta = 0$  |
| Charge dispersion        | Large (sensitive to $n_g$ ) | Exponentially small              |
| Anharmonicity            | Large ( $\sim E_C$ )        | Small ( $ \alpha  = E_C/\hbar$ ) |

## 4. The Transmon: Charge-Noise Insensitive CPB

### 4.1 The Key Insight of Koch et al. (2007)

The transmon is *not* a new circuit — it is the *same* CPB Hamiltonian  $H = 4E_C(\hat{n} - n_g)^2 - E_J \cos \hat{\delta}$ , simply operated at  $E_J/E_C \gg 1$  (typically 30–100). The crucial discovery: as  $E_J/E_C$  increases,

- The **charge dispersion** (sensitivity to  $n_g$ ) **decreases exponentially**:  $\epsilon_m \propto e^{-\sqrt{8E_J/E_C}}$ .
- **The anharmonicity decreases** only *algebraically*:  $\alpha \sim -E_C$ .

So there is a wide window of  $E_J/E_C$  where charge noise is completely suppressed while sufficient anharmonicity remains for qubit operation. This is the transmon regime.

The dephasing time from charge noise scales as  $T_2 \sim \hbar/(e^2\pi|\epsilon_1|)$ , which rises exponentially with  $E_J/E_C$ , reaching potentially above seconds.

### 4.2 Deriving the Transmon Hamiltonian

#### From CPB to anharmonic oscillator via Taylor expansion

For  $E_J/E_C \gg 1$ , the phase fluctuations are small ( $\delta \approx 0$ ). Expand the cosine:

$$-E_J \cos \delta = -E_J + \frac{E_J}{2} \delta^2 - \frac{E_J}{24} \delta^4 + \mathcal{O}(\delta^6) \quad (16)$$

Dropping the constant  $-E_J$  and keeping to 4th order:

$$\hat{H} \approx 4E_C \hat{n}^2 + \frac{E_J}{2} \hat{\delta}^2 - \frac{E_J}{24} \hat{\delta}^4 \quad (17)$$

(We dropped  $n_g$  because charge dispersion is exponentially suppressed.)

**Step 1: Identify the harmonic part.** The quadratic terms  $4E_C \hat{n}^2 + \frac{E_J}{2} \hat{\delta}^2$  form a harmonic oscillator. Using  $[\hat{\delta}, \hat{n}] = i$  and defining:

$$\omega_p = \sqrt{8E_J E_C} / \hbar \quad (\text{plasma frequency}) \quad (18)$$

we introduce ladder operators  $\hat{a}, \hat{a}^\dagger$  such that:

$$\hat{\delta} = \delta_{\text{zpf}} (\hat{a} + \hat{a}^\dagger), \quad \hat{n} = -in_{\text{zpf}} (\hat{a} - \hat{a}^\dagger) \quad (19)$$

with  $\delta_{\text{zpf}} = (2E_C/E_J)^{1/4}$  and  $n_{\text{zpf}} = (E_J/(2E_C))^{1/4} \cdot \frac{1}{2}$ .

The harmonic part becomes  $\hbar\omega_p \hat{a}^\dagger \hat{a}$  (dropping the zero-point energy).

**Step 2: Express the quartic perturbation in ladder operators.**

$$-\frac{E_J}{24} \hat{\delta}^4 = -\frac{E_J}{24} \delta_{\text{zpf}}^4 (\hat{a} + \hat{a}^\dagger)^4 = -\frac{E_J}{24} \left( \frac{2E_C}{E_J} \right) (\hat{a} + \hat{a}^\dagger)^4 = -\frac{E_C}{12} (\hat{a} + \hat{a}^\dagger)^4 \quad (20)$$

**Step 3: Apply the rotating-wave approximation (RWA) to  $(\hat{a} + \hat{a}^\dagger)^4$ .**

Expand  $(\hat{a} + \hat{a}^\dagger)^4$  and keep only the terms that conserve excitation number (the “secular” terms). The non-secular terms like  $\hat{a}^4$ ,  $(\hat{a}^\dagger)^4$ ,  $\hat{a}^\dagger \hat{a}^\dagger \hat{a}^\dagger \hat{a}$ , etc. oscillate rapidly and average to zero. The surviving terms are:

$$(\hat{a} + \hat{a}^\dagger)^4 \xrightarrow{\text{RWA}} 6\hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} + 6\hat{a}^\dagger \hat{a} + 3 = 6(\hat{a}^\dagger \hat{a})^2 + 6\hat{a}^\dagger \hat{a} + 3 \quad (21)$$

(This uses  $\langle n | (\hat{a} + \hat{a}^\dagger)^4 | n \rangle = 6n^2 + 6n + 3$ , which counts all Wick contractions.)

But we must be careful with operator ordering. Using normal ordering:

$$(\hat{a} + \hat{a}^\dagger)^4 \xrightarrow{\text{RWA}} 6\hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} + 12\hat{a}^\dagger \hat{a} + 3 \quad (22)$$

So:

$$\begin{aligned} -\frac{E_C}{12}(\hat{a} + \hat{a}^\dagger)^4 &\xrightarrow{\text{RWA}} -\frac{E_C}{12}(6\hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a} + 12\hat{a}^\dagger\hat{a} + 3) \\ &= -\frac{E_C}{2}\hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a} - E_C\hat{a}^\dagger\hat{a} - \frac{E_C}{4} \end{aligned} \quad (23)$$

**Step 4: Combine with the harmonic part.**

$$\begin{aligned} \hat{H} &= \hbar\omega_p \hat{a}^\dagger\hat{a} - \frac{E_C}{2}\hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a} - E_C\hat{a}^\dagger\hat{a} + \text{const.} \\ &= (\hbar\omega_p - E_C)\hat{a}^\dagger\hat{a} - \frac{E_C}{2}\hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a} + \text{const.} \end{aligned} \quad (24)$$

Defining the **qubit frequency**  $\hbar\omega_q = \hbar\omega_p - E_C = \sqrt{8E_J E_C} - E_C$  and the **anharmonicity**  $\alpha = -E_C/\hbar$ :

$$\boxed{\frac{\hat{H}}{\hbar} = \omega_q \hat{a}^\dagger\hat{a} + \frac{\alpha}{2} \hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a}} \quad (25)$$

The energy of level  $|n\rangle$  is  $E_n/\hbar = \omega_q n + \frac{\alpha}{2}n(n-1)$ , giving transition frequencies:

$$\omega_{n,n+1} = \omega_q + \alpha n \quad (26)$$

### 4.3 Two-Level Approximation: The Qubit

Truncating to the  $\{|0\rangle, |1\rangle\}$  subspace, the  $\hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a}$  term vanishes (it requires at least 2 excitations), giving:

$$\hat{a}^\dagger\hat{a} \rightarrow \frac{1-\sigma_z}{2}, \quad \hat{a} \rightarrow \sigma^- = |0\rangle\langle 1|, \quad \hat{a}^\dagger \rightarrow \sigma^+ = |1\rangle\langle 0| \quad (27)$$

$$\boxed{\frac{\hat{H}}{\hbar} = -\frac{\omega_q}{2}\sigma_z} \quad (28)$$

This is the free qubit Hamiltonian — a spin-1/2 in an effective magnetic field along  $z$ .



## 5. Inductively Shunted Qubits

### 5.1 Motivation: A Different Route to Charge Insensitivity

The transmon achieves charge insensitivity by making  $E_J/E_C \gg 1$ , at the cost of reduced anharmonicity. An alternative approach: **shunt the junction with an inductor**. This fundamentally changes the topology of the circuit.

### 5.2 The Circuit and its Hamiltonian

A Josephson junction (with self-capacitance  $C$ ) in parallel with an inductor  $L$ , biased by an external flux  $\phi_s$  through the loop.

#### Deriving the inductively shunted qubit Hamiltonian

Using the circuit quantization recipe with one active node:

- **Kinetic energy** (capacitive):  $T = \frac{C}{2} \dot{\phi}^2$
- **Potential energy: Josephson + inductive**. The inductor is a closure branch forming a loop with the junction. By flux quantization, if the junction has flux  $\phi$  and the inductor has flux  $\phi_L$ , KVL around the loop gives  $\phi + \phi_L = \phi_s$  (external flux). So  $\phi_L = \phi - \phi_s$  (depending on sign convention). The inductive energy is  $\frac{1}{2L}(\phi - \phi_s)^2$ .

The Lagrangian:

$$\mathcal{L} = \frac{C}{2} \dot{\phi}^2 + E_J \cos \frac{\phi}{\varphi_0} - \frac{1}{2L}(\phi - \phi_s)^2 \quad (29)$$

Legendre transform ( $Q = C\dot{\phi}$ ):

$$H = \frac{Q^2}{2C} - E_J \cos \frac{\phi}{\varphi_0} + \frac{1}{2L}(\phi - \phi_s)^2 \quad (30)$$

In dimensionless variables  $\delta = \phi/\varphi_0$ ,  $\hat{n} = Q/(2e)$ ,  $\delta_s = \phi_s/\varphi_0$ :

$$\hat{H} = E_C \hat{n}^2 - E_J \cos \hat{\delta} + E_L (\hat{\delta} - \delta_s)^2 \quad (31)$$

where  $E_C = (2e)^2/(2C)$ ,  $E_L = \varphi_0^2/(2L)$ , and  $\delta_s = \phi_s/\varphi_0$ .

### 5.3 The Rich Parameter Landscape

This Hamiltonian has **four parameters**:  $E_C$ ,  $E_J$ ,  $E_L$ , and  $\delta_s$ . Different regimes give different qubit types:

#### Inductively Shunted Qubit Zoo

- **Phase qubit**:  $E_L \ll E_J$ , large  $E_J/E_C$ . The inductor provides a small tilt to the cosine, creating a “washboard” potential with metastable wells. The qubit lives in 2–3 levels of one well.
- **Flux qubit**:  $E_J > E_L$ , moderate  $E_J/E_C$ ,  $\delta_s \approx \pi$ . At half flux quantum, the potential has a symmetric double well. The qubit states are  $|\odot\rangle \pm |\oslash\rangle$  (clockwise/counterclockwise persistent current superpositions).
- **Fluxonium**:  $E_L \ll E_J$ ,  $E_C$  comparable, large superinductance  $L$ . Combines charge insensitivity (from the inductor shunt, which makes  $\delta$  non-compact) with large anharmonicity. Currently among the highest-coherence superconducting qubits.

## 5.4 Charge Noise Insensitivity — CPB vs. Inductively Shunted

### Critical Point

This is a **topological** distinction, not just a quantitative one.

**CPB (no shunt inductor):** The island is **electrically isolated**. Cooper pairs can only reach the island through the junction, so  $\hat{n}$  is *integer-valued*. The phase  $\delta$  is a **compact variable** (defined on  $[0, 2\pi)$  with periodic boundary conditions:  $|\delta\rangle = |\delta + 2\pi\rangle$ ). The offset charge  $n_g$  enters the Hamiltonian and *cannot be screened* — it represents real stray charges trapped on the island.

**Inductively shunted:** The island is connected to ground through the inductor. Cooper pairs can flow freely onto the island through this DC path, so  $\hat{n}$  becomes a *continuous* variable. The phase  $\delta$  becomes a **non-compact variable** (extended,  $\delta \in (-\infty, \infty)$ ) because the  $E_L \hat{\delta}^2$  term breaks the  $2\pi$  periodicity ( $|\delta\rangle \neq |\delta + 2\pi\rangle$  — different phases correspond to different currents in the inductor). The offset charge  $n_g$  *can be screened* through the inductor — it doesn't appear in the spectrum.

## 6. Circuit Quantum Electrodynamics (cQED)

### 6.1 The Setup

Circuit QED is the solid-state analogue of cavity QED: a “two-level atom” (the transmon qubit) interacts with a single mode of the electromagnetic field (a superconducting resonator) via the electric field (capacitive coupling).

### 6.2 The Jaynes–Cummings Hamiltonian from Circuit Elements

Starting from the coupled circuit (derived in detail in our previous worked example):

$$\frac{\hat{H}}{\hbar} = -\frac{\omega_q}{2}\hat{\sigma}_z + \omega_r\hat{a}^\dagger\hat{a} + g(\hat{\sigma}^- - \hat{\sigma}^+)(\hat{a} - \hat{a}^\dagger) \quad (32)$$

where  $g = \frac{4e^2C_g}{C_1C_2}n_{1,\text{zpf}}n_{2,\text{zpf}}$  involves the zero-point charge fluctuations of both the qubit and resonator, mediated by the coupling capacitor  $C_g$ .

The voltage operator for each mode is  $\hat{V} = -iV_{\text{zpf}}(\hat{a} - \hat{a}^\dagger) = -i\frac{2e}{C}n_{\text{zpf}}(\hat{a} - \hat{a}^\dagger)$ , so the interaction  $H_{\text{int}} = C_g\hat{V}_1\hat{V}_2$  produces the  $(\hat{\sigma}^- - \hat{\sigma}^+)(\hat{a} - \hat{a}^\dagger)$  coupling.

Expanding:

$$(\hat{\sigma}^- - \hat{\sigma}^+)(\hat{a} - \hat{a}^\dagger) = \hat{\sigma}^-\hat{a} - \hat{\sigma}^-\hat{a}^\dagger - \hat{\sigma}^+\hat{a} + \hat{\sigma}^+\hat{a}^\dagger \quad (33)$$

Applying the **RWA** (dropping  $\hat{\sigma}^-\hat{a}$  and  $\hat{\sigma}^+\hat{a}^\dagger$  which oscillate at  $\pm(\omega_q + \omega_r)$ ):

#### Key Result

**Jaynes–Cummings Hamiltonian:**

$$\frac{\hat{H}}{\hbar} = -\frac{\omega_q}{2}\hat{\sigma}_z + \omega_r\hat{a}^\dagger\hat{a} + g(\hat{\sigma}^-\hat{a}^\dagger + \hat{\sigma}^+\hat{a}) \quad (34)$$

Coupling strength:  $g = \frac{4e^2C_g}{C_1C_2}$ , giving  $\nu_{\text{vac}} = g/\pi \approx 1\text{--}300$  MHz.

The condition  $g \gg [\kappa, \gamma]$  (coupling much larger than **photon decay  $\kappa$**  and **qubit decay  $\gamma$** ) defines the **strong coupling regime**, achievable in circuit QED.

## 7. Coupled Systems: General Dynamics

### 7.1 Two Coupled Two-Level Systems

When both the qubit and the resonator can be treated as two-level systems (valid for the lowest excitation sector), the interaction becomes:

$$H_{\text{int}} = \hbar g (\sigma_1^- - \sigma_1^+) (\sigma_2^- - \sigma_2^+) \quad (35)$$

The full Hamiltonian in the basis  $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$  (where  $|q, r\rangle$  labels qubit and resonator states):

$$\mathbf{H} = \hbar \begin{pmatrix} \Delta & 0 & 0 & 0 \\ 0 & \Delta & 2g & 0 \\ 0 & 2g & -\Delta & 0 \\ 0 & 0 & 0 & -\Delta \end{pmatrix} \quad (36)$$

where  $\Delta = (\omega_1 - \omega_2)/2$  is the half-detuning.

#### Diagonalization in the single-excitation subspace

The states  $|00\rangle$  and  $|11\rangle$  are eigenstates (no coupling to other states in RWA). The interesting physics is in the  $\{|01\rangle, |10\rangle\}$  subspace:

$$H_{\text{lexc}} = \hbar \begin{pmatrix} \Delta & 2g \\ 2g & -\Delta \end{pmatrix} = \frac{\hbar\Delta}{2} \tilde{\sigma}_z + \hbar g \tilde{\sigma}_x \quad (37)$$

This is *mathematically identical* to a spin-1/2 in a magnetic field  $\vec{M} = (2g, 0, \Delta)$ . The “field direction” makes angle  $\theta_M$  with the  $z$ -axis, where  $\tan \theta_M = 2g/\Delta$ .

**Eigenvalues:**

$$E_{\pm} = -\frac{\hbar\omega_1}{2} \pm \frac{\hbar}{2} \sqrt{4g^2 + \Delta^2} \quad (38)$$

**Eigenstates (dressed states):**

$$|+\rangle = \sin \frac{\theta_M}{2} |01\rangle + \cos \frac{\theta_M}{2} |10\rangle \quad (39)$$

$$|-\rangle = \sin \frac{\theta_M}{2} |01\rangle - \cos \frac{\theta_M}{2} |10\rangle \quad (40)$$

**Resonant case** ( $\Delta = 0, \theta_M = \pi/2$ ):  $|+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, |-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$  with splitting  $2g$  (vacuum Rabi splitting).

**Dispersive limit** ( $|\Delta| \gg g, \theta_M \rightarrow 0$  or  $\pi$ ):  $|+\rangle \approx |10\rangle - \frac{g}{\Delta} |01\rangle, |-\rangle \approx -|01\rangle - \frac{g}{\Delta} |10\rangle$ . The states are weakly dressed.

## 8. Dispersive Readout

### 8.1 The Dispersive Hamiltonian

In the dispersive regime  $|\Delta| = |\omega_q - \omega_r| \gg g$ , the Jaynes–Cummings Hamiltonian can be approximately diagonalized (via the **Schrieffer–Wolff** / Fröhlich transformation).

For the multi-photon case, the dressed states at photon number  $n$  are:

$$|n, +\rangle = -\frac{g\sqrt{n}}{\Delta}|n, 1\rangle + |n+1, 0\rangle \quad (41)$$

$$|n, -\rangle = -|n, 1\rangle - \frac{g\sqrt{n}}{\Delta}|n+1, 0\rangle \quad (42)$$

with energies  $E_{\pm} = n\hbar\omega_r \pm \frac{\hbar}{2}\sqrt{4g^2(n+1) + \Delta^2}$ .

The effective dispersive Hamiltonian (including the correction from the transmon's third level):

#### Key Result

**Dispersive Hamiltonian:**

$$\frac{\hat{H}}{\hbar} = (\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a} + \frac{\tilde{\omega}_q}{2}\hat{\sigma}_z \quad (43)$$

with  $\tilde{\omega}_q = \omega_q + \chi$  (Lamb shift) and  $\chi = g^2/\Delta$  (dispersive shift).

For a transmon with anharmonicity  $\alpha$ , the more accurate expression is:

$$\chi \approx -\frac{g^2\alpha}{\Delta(\Delta + \alpha)} \quad (44)$$

### 8.2 Physical Meaning: Qubit-State-Dependent Resonator Frequency

The resonator frequency depends on the qubit state:

$$\text{Qubit in } |g\rangle : \omega_r^{(g)} = \omega_r - \chi \quad (45)$$

$$\text{Qubit in } |e\rangle : \omega_r^{(e)} = \omega_r + \chi \quad (46)$$

By sending a probe tone at  $\omega_r$  and measuring the transmitted/reflected signal, the *phase and amplitude* of the output depend on whether the resonator is at  $\omega_r - \chi$  or  $\omega_r + \chi$ . This implements a **quantum non-demolition (QND) measurement**: reading out  $\sigma_z$  without causing transitions.

Conversely, the qubit sees a photon-number-dependent frequency:  $\tilde{\omega}_q = \omega_q + \chi + 2\chi\hat{a}^\dagger\hat{a}$  (**AC Stark shift** / photon number splitting,  $2\chi$  per photon).

## 9. Qubit Drive

A microwave drive at frequency  $\omega_d$  is applied through a drive capacitor  $C_d$ :

$$H_d = C_d V_1 V_d(t) = -i\Omega_d V_d(t)(\hat{a} - \hat{a}^\dagger) \quad (47)$$

where  $\Omega_d = \frac{2eC_d}{C_\Sigma} n_{\text{zpf}}$  and  $V_d(t) = V_0 s(t) \sin(\omega_d t + \phi)$ .

After RWA:

$$H_d = -\Omega_d V_0 s(t) (\hat{a} e^{i(\omega_d t + \phi)} - \hat{a}^\dagger e^{-i(\omega_d t + \phi)}) \quad (48)$$

Absorbing the phase into the envelope  $\xi(t) = s(t)e^{-i\phi}$ :

$$H_d = -\Omega_d V_0 (\hat{a} \xi^* e^{i\omega_d t} - \hat{a}^\dagger \xi e^{-i\omega_d t}) \quad (49)$$

This is equivalent to coupling to a classical coherent field with amplitude  $V_0$  (replacing  $\sqrt{n+1} \rightarrow V_0$  in the Jaynes–Cummings model). In the rotating frame at  $\omega_d$ , the qubit drive Hamiltonian becomes the standard Rabi Hamiltonian used for single-qubit gates.

## 10. Decoherence and Dissipation

### 10.1 The General Framework

A qubit coupled to its environment experiences two types of decoherence:

1. **Relaxation (energy decay,  $T_1$ )**: The qubit spontaneously decays from  $|1\rangle$  to  $|0\rangle$  by emitting energy into the environment. On the Bloch sphere, this corresponds to the state vector shrinking toward the north pole ( $|0\rangle$ ).

2. **Pure dephasing ( $T_\phi$ )**: Low-frequency noise in the qubit's Hamiltonian parameters (gate charge  $n_g$ , flux  $\Phi$ , etc.) causes the qubit frequency to fluctuate randomly, leading to loss of phase coherence. On the Bloch sphere, this corresponds to diffusion around the  $z$ -axis.

The total decoherence rate  $\Gamma_2 = 1/T_2$  combines both:

$$\frac{1}{T_2} = \Gamma_2 = \frac{\Gamma_1}{2} + \Gamma_\phi \quad (50)$$

### 10.2 Relaxation: Fermi's Golden Rule for Circuits

#### $T_1$ from the noise spectral density

The qubit Hamiltonian depends on some parameter  $\lambda$  (e.g., gate charge, flux, critical current). Fluctuations  $\delta\lambda(t)$  couple to the qubit:

$$\hat{H}_q = -\frac{1}{2} \hat{\sigma} \cdot \left( \vec{H}_0(\lambda_0) + \frac{\partial \vec{H}_0}{\partial \lambda} \delta\lambda + \dots \right) \quad (51)$$

The transverse (off-diagonal) part of this coupling causes transitions between  $|0\rangle$  and  $|1\rangle$ . Define the **transverse coupling matrix element**:

$$D_{\lambda,\perp} = \frac{1}{\hbar} \langle 0 | \frac{\partial H}{\partial \lambda} | 1 \rangle \quad (52)$$

By Fermi's golden rule, the relaxation rate is:

$$\Gamma_1 = T_1^{-1} = \frac{\pi}{2} D_{\lambda,\perp}^2 S_\lambda(\omega_{01}) \quad (53)$$

where  $S_\lambda(\omega_{01})$  is the **spectral density of the noise** in parameter  $\lambda$ , evaluated at the qubit transition frequency  $\omega_{01}$ .

### 10.3 Pure Dephasing: Sensitivity to Low-Frequency Noise

#### $T_\phi$ from the longitudinal noise

The longitudinal (diagonal) part of the noise modulates the qubit frequency:  $\omega_{01}(\lambda) = \omega_{01}(\lambda_0) + \frac{\partial \omega_{01}}{\partial \lambda} \delta\lambda + \dots$

This causes the relative phase between  $|0\rangle$  and  $|1\rangle$  to accumulate randomly:  $\phi(t) = \int_0^t \delta\omega_{01}(t') dt'$ . The decay of the off-diagonal density matrix element is:

$$f(\phi) \equiv \langle e^{i\phi(t)} \rangle \approx e^{-\Gamma_2 t} \quad (54)$$

Define the **longitudinal coupling**:

$$D_{\lambda,\parallel}^2 = \left( \frac{\partial \omega_{01}}{\partial \lambda} \right)^2 \quad (55)$$

The pure dephasing rate is:

$$\Gamma_\phi = \pi D_{\lambda,\parallel}^2 S_\lambda(0) \quad (56)$$

where  $S_\lambda(0)$  is the noise spectral density at *zero frequency* (low-frequency fluctuations). This is why pure dephasing is dominated by  $1/f$  noise (which diverges at low frequency).

### Physical Insight

The key design principle for long coherence:

- To minimize  $T_1^{-1}$ : reduce  $D_{\lambda,\perp}$  (the off-diagonal matrix element) and/or reduce  $S_\lambda(\omega_{01})$  (the noise at the qubit frequency). The latter means avoiding lossy materials (dielectric loss, vortex loss, quasiparticles).
- To minimize  $\Gamma_\phi$ : operate at a **sweet spot** where  $\frac{\partial \omega_{01}}{\partial \lambda} = 0$ , making  $D_{\lambda,\parallel} = 0$  to first order. For the CPB,  $n_g = 0.5$  is a charge sweet spot. For the transmon,  $\frac{\partial \omega_{01}}{\partial n_g} \approx 0$  everywhere (exponential suppression). For flux qubits, the sweet spot is at half flux quantum  $\Phi_{\text{ext}} = \phi_0/2$ .

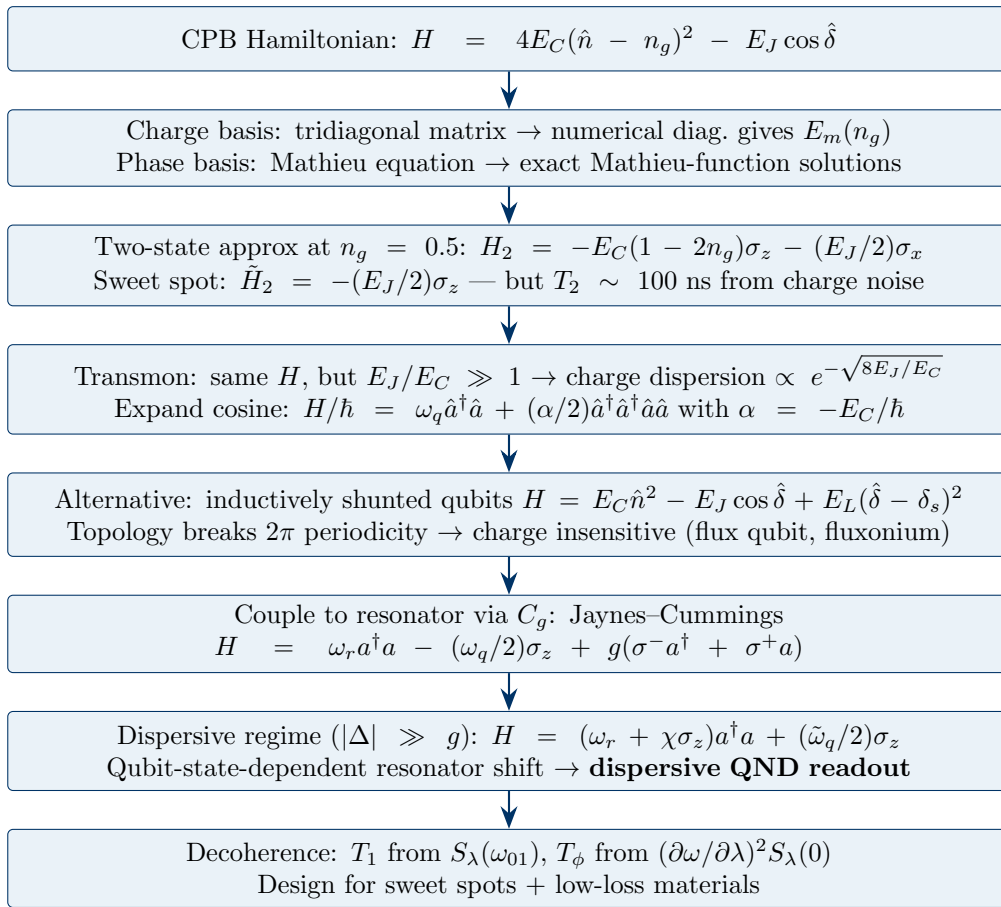
## 10.4 Sources of Noise

| Noise source                  | Type               | Affects    | CPB $T_2$          | Transmon $T_2$            |
|-------------------------------|--------------------|------------|--------------------|---------------------------|
| Charge noise ( $\delta n_g$ ) | Low freq ( $1/f$ ) | Dephasing  | $\sim 100$ ns      | $\sim 100$ ms             |
| Flux noise ( $\delta \Phi$ )  | Low freq ( $1/f$ ) | Dephasing  | $\sim 100$ $\mu$ s | $\sim 100$ $\mu$ s        |
| Dielectric loss               | High freq          | Relaxation | Limits $T_1$       | $\sim 10$ – $500$ $\mu$ s |
| Quasiparticles                | High freq          | Relaxation | Limits $T_1$       | sporadically              |
| Magnetic vortices             | High freq          | Relaxation | Limits $T_1$       | in thin films             |

The transmon's exponential suppression of charge dispersion pushed  $T_2$  from charge noise from nanoseconds to potentially seconds, making flux noise and dielectric loss the new bottleneck. Current state-of-the-art transmons achieve  $T_1 \sim 100$ – $500$   $\mu$ s and  $T_2 \sim 100$ – $300$   $\mu$ s.



## 11. Summary: The Complete Logical Chain



## 12. Essential Formulas Reference

### 12.1 Cooper Pair Box / Transmon

$$H_{\text{CPB}} = 4E_C(\hat{n} - n_g)^2 - E_J \cos \hat{\delta} \quad E_C = e^2/(2C_\Sigma), \quad n_g = C_g V_g/(2e) \quad (57)$$

$$\omega_q = \sqrt{8E_J E_C}/\hbar - E_C/\hbar \quad \alpha = -E_C/\hbar \quad (58)$$

$$H_{\text{transmon}}/\hbar = \omega_q \hat{a}^\dagger \hat{a} + (\alpha/2) \hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} \quad \omega_{n,n+1} = \omega_q + n\alpha \quad (59)$$

### 12.2 Inductively Shunted Qubit

$$H = E_C \hat{n}^2 - E_J \cos \hat{\delta} + E_L (\hat{\delta} - \delta_s)^2 \quad E_L = \varphi_0^2/(2L), \quad \delta_s = \phi_s/\varphi_0 \quad (60)$$

### 12.3 Jaynes–Cummings / cQED

$$H/\hbar = -(\omega_q/2)\sigma_z + \omega_r \hat{a}^\dagger \hat{a} + g(\sigma^- \hat{a}^\dagger + \sigma^+ \hat{a}) \quad g = 4e^2 C_g/(C_1 C_2) \quad (61)$$

$$\text{Dressed states: } E_\pm = n\hbar\omega_r \pm (\hbar/2)\sqrt{4g^2(n+1) + \Delta^2} \quad \Delta = \omega_q - \omega_r \quad (62)$$

### 12.4 Dispersive Regime

$$H_{\text{disp}}/\hbar = (\omega_r + \chi\sigma_z)\hat{a}^\dagger \hat{a} + (\tilde{\omega}_q/2)\sigma_z \quad \chi = g^2/\Delta \quad (63)$$

$$\tilde{\omega}_q = \omega_q + \chi \text{ (Lamb shift)} \quad \chi_{\text{transmon}} \approx -g^2\alpha/[\Delta(\Delta + \alpha)] \quad (64)$$

### 12.5 Decoherence

$$\Gamma_1 = (\pi/2)D_{\lambda,\perp}^2 S_\lambda(\omega_{01}) \quad D_{\lambda,\perp} = \hbar^{-1}\langle 0|\partial H/\partial \lambda|1\rangle \quad (65)$$

$$\Gamma_\phi = \pi(\partial\omega_{01}/\partial\lambda)^2 S_\lambda(0) \quad \Gamma_2 = \Gamma_1/2 + \Gamma_\phi \quad (66)$$

*References:* Koch et al., PRA 76, 042319 (2007); Blais et al., PRA 69, 062320 (2004); Bouchiat et al., Physica Scripta T76, 165 (1998); Mooij et al., Science 285 (1999); Ithier et al., PRB 72, 134519 (2005); Vool & Devoret, arXiv:1610.03438; Kjaergaard et al., Ann. Rev. Cond. Mat. Phys. 11 (2020).